Mecanum Robot Digital Twin — Complete Project Handoff v2

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1. Project Objective

Build a real-time digital twin of a mecanum wheel AMR using NVIDIA Isaac Sim 5.1. Physical robot streams encoder, IMU, and camera data via micro-ROS and ROS 2. Long-term goal: RL training with Isaac Lab + sim-to-real transfer.

Phase	Description	Status
1	Physical robot — micro-ROS, encoders, IMU, EKF, RViz	COMPLETE
2	Isaac Sim digital twin via ROS 2 bridge + FastDDS	COMPLETE
3A	Jetson Orin Nano + OAK-D — ROS 2 perception stack	COMPLETE
3B	Obstacle avoidance using OAK-D depth + Nav2	IN PROGRESS
4	Nav2 in simulation, then real robot	PENDING
5	Isaac Lab RL training + sim-to-real deployment	PENDING
6	Fix mecanum roller physics (individual roller colliders)	PENDING

2. Hardware Configuration

Physical Robot

Component	Specification
Chassis	goBILDA Strafer Chassis Kit v4
Microcontroller	Arduino Portenta H7 + Mid Carrier
Motor Controllers	Dual RoboClaw 2x15A (RC1: 0x80, RC2: 0x81)
Motors	goBILDA 5203 (19.2:1, 312 RPM)
Wheels	goBILDA Mecanum 96mm (2 LH + 2 RH) — Radius 48mm
Encoder CPR	2150 counts/revolution
IMU	SparkFun ISM330DHCX (I2C 0x6B)
Chassis Mass	4.966 kg | Wheel Mass: 0.207 kg each
Communication	micro-ROS over WiFi UDP port 8888
WiFi	SSID: PortentaROS | Password: microros123 | Robot IP: 10.42.0.128

Compute & Perception Hardware

Device	Specs	Role	IP
Ubuntu Laptop	RTX 3070 / Ubuntu 22.04	ROS 2 master, EKF, RViz, micro-ROS agent	10.42.0.1
Windows Desktop	RTX 5080 / Windows 11	Isaac Sim 5.1 digital twin	10.42.0.71

Jetson Orin Nano	8GB / Ubuntu 22.04 / JetPack 6.2.2	OAK-D perception + ROS 2 node	WiFi DHCP
OAK-D (Luxonis)	4 TOPS NPU / Stereo depth	Depth camera on robot	USB-C to Jetson

3. Software Stack

Ubuntu Laptop (10.42.0.1)

Component	Details
OS	Ubuntu 22.04 LTS
ROS 2	Humble Hawksbill
micro-ROS Agent	Docker — microros/micro-ros-agent:humble
EKF	robot_localization ekf_node
Key Scripts	joint_state_fix.py, digital_twin.py, odom_bridge.py, teleop_qweasdzxc.py

Jetson Orin Nano

Component	Details
OS	Ubuntu 22.04 LTS (JetPack 6.2.2)
Storage	WD Black SN750 500GB NVMe (flashed via SDK Manager)
ROS 2	Humble Hawksbill
Camera Driver	depthai + ros-humble-depthai-ros
Topics published	/oak/rgb/image_raw, /oak/stereo/image_raw, /oak/nn/spatial_detections, /pointcloud

Windows Desktop (10.42.0.71)

Component	Details
OS	Windows 11 (25H2)
GPU Driver	580.88 WHQL — DO NOT UPDATE (595.79 breaks Isaac Sim)
Isaac Sim	5.1.0 standalone at C:\isaacsim\isaac-sim-standalone-5.1.0-windows-x86_64\
USD Scene	C:\isaac-files\robot_v1_windows.usd
FastDDS Config	C:\Users\ricar\fastdds_config.xml (peer: 10.42.0.1)

4. Network Configuration

Device	IP	Connection
Ubuntu Laptop	10.42.0.1	Hotspot host (PortentaROS)
Windows Desktop	10.42.0.71	WiFi client to PortentaROS
Portenta H7	10.42.0.128	WiFi client — micro-ROS UDP 8888
Jetson Orin Nano	DHCP	WiFi client (home WiFi or PortentaROS)

Note: Jetson gets better pointcloud Hz (~2.5) on home WiFi vs PortentaROS hotspot (~0.4 Hz). Keep Jetson on home WiFi. Portenta must stay on PortentaROS hotspot for micro-ROS agent discovery.

5. ROS 2 Topics

Topic	Type	Flow
/cmd_vel	geometry_msgs/Twist	teleop → Portenta H7 → motors
/joint_states	sensor_msgs/JointState	Portenta H7 → Ubuntu
/joint_states_fixed	sensor_msgs/JointState	joint_state_fix.py → Isaac Sim (FR/RR inverted, SCALE=6.0)
/odom	nav_msgs/Odometry	Portenta H7 → Ubuntu
/odometry/filtered	nav_msgs/Odometry	EKF → RViz + Isaac Sim
/imu	sensor_msgs/Imu	Portenta H7 → Ubuntu
/oak/rgb/image_raw	sensor_msgs/Image	Jetson OAK-D → ROS 2 network
/oak/stereo/image_raw	sensor_msgs/Image	Jetson OAK-D → ROS 2 network
/oak/nn/spatial_detections	depthai_ros_msgs/SpatialDetectionArray	Jetson OAK-D → ROS 2 network
/pointcloud	sensor_msgs/PointCloud2	Jetson (depth_image_proc) → Ubuntu → RViz

6. Startup Sequences

Ubuntu Laptop — 8 Terminals

Terminal	Command
T1 — Hotspot	nmcli connection up Hotspot
T2 — micro-ROS	docker run -it --rm --net=host microros/micro-ros-agent:humble udp4 --port 8888 -v4
T3 — RSP	source /opt/ros/humble/setup.bash && ros2 run robot_state_publisher robot_state_publisher --ros-args -p robot_description:=
T4 — EKF	source /opt/ros/humble/setup.bash && ros2 run robot_localization ekf_node --ros-args --params-file ~/mecanum_viz_ws/src/
T5 — Twin	source /opt/ros/humble/setup.bash && python3 ~/digital_twin.py
T6 — Fix	source /opt/ros/humble/setup.bash && python3 ~/joint_state_fix.py
T7 — RViz	source /opt/ros/humble/setup.bash && ros2 run rviz2 rviz2
T8 — Teleop	source /opt/ros/humble/setup.bash && python3 ~/Desktop/teleop_qweasdzxc.py

Jetson Orin Nano — 2 Terminals

Terminal	Command
T1 — OAK-D Driver	source /opt/ros/humble/setup.bash && ros2 launch depthai_ros_driver camera.launch.py
T2 — PointCloud	source /opt/ros/humble/setup.bash && ros2 run depth_image_proc point_cloud_xyz_node --ros-args --remap image_r

Windows — Isaac Sim

```
C:\isaacsim\isaac-sim-standalone-5.1.0-windows-x86_64\isaac-sim.bat
--/isaac/startup/ros_bridge_extension=isaacsim.ros2.bridge
```

7. RViz Pointcloud Configuration

To visualize the OAK-D pointcloud in RViz on the laptop:

Setting	Value
Fixed Frame	oak_rgb_camera_optical_frame
Display Type	PointCloud2
Topic	/pointcloud
Reliability Policy	Best Effort ← CRITICAL
Style	Flat Squares
Size (m)	0.01

■ Reliability Policy MUST be Best Effort — default Reliable will show 0 points due to QoS mismatch.

8. Isaac Sim Setup

Parameter	Value
USD Scene	C:\isaac-files\robot_v1_windows.usd
Articulation Root	/robot_m/base_link
Joint Names	fl_wheel_joint, fr_wheel_joint, rl_wheel_joint, rr_wheel_joint
Angular Drive Stiffness	0.0
Angular Drive Damping	10000.0
Angular Drive Max Force	100000.0
Action Graph Input	/joint_states_fixed
Articulation Controller Target	/robot_m

9. Phase 3B — Obstacle Avoidance (Next Steps)

Goal: Use OAK-D depth data on the Jetson to detect obstacles and stop/avoid them autonomously.

Step	Task	Tool
3B-1	Install Nav2 on Ubuntu laptop	sudo apt install ros-humble-navigation2 ros-humble-nav2-bringup
3B-2	Create costmap from /pointcloud	Nav2 local costmap with PointCloud2 sensor source
3B-3	Add obstacle layer to costmap	nav2_costmap_2d ObstacleLayer
3B-4	Test stop-on-obstacle behavior	Send goal via RViz 2D Nav Goal
3B-5	Tune inflation radius and obstacle range	nav2_params.yaml
3B-6	Integrate with Isaac Sim	Subscribe /cmd_vel from Nav2 in Isaac Sim

10. Key Files

File	Location	Description
URDF	~/mecanum_viz_ws/src/mecanum_description/urdf/mecanum_robot.urdf	Robot description
EKF config	~/mecanum_viz_ws/src/mecanum_description/config/ekf.yaml	EKF parameters

digital_twin.py	~/digital_twin.py	Publishes TF + joint states for RViz
joint_state_fix.py	~/joint_state_fix.py	Mirrors FR/RR joints, SCALE=6.0
odom_bridge.py	~/odom_bridge.py	Optional: pose to Isaac Sim
teleop	~/Desktop/teleop_qweasdzxc.py	Keyboard teleop QWEASDZXC
FastDDS XML	C:\Users\ricar\fastdds_config.xml	Peer: 10.42.0.1
USD Scene	C:\isaac-files\robot_v1_windows.usd	Isaac Sim scene with physics
RViz config	~/oak_pointcloud.rviz	OAK-D pointcloud visualization

11. Critical Warnings

■ NEVER update NVIDIA driver above 580.88 on Windows desktop — 595.79 crashes Isaac Sim

■ NEVER save Isaac Sim with odom_bridge.py running — modifies prim transforms at runtime

■ joint_state_fix.py SCALE=6.0 is calibrated — changing it affects Isaac Sim sync

■ Jetson Orin Nano needs 7-20V via barrel jack — standard 5V powerbank will NOT work

■ PortentaROS hotspot password: microros123

■ Isaac Sim must launch with --/isaac/startup/ros_bridge_extension=isaacsim.ros2.bridge flag

■ USD assets at C:\isaac-files\3-18-2026\USD\ — do NOT move these files

■ RViz PointCloud2 Reliability Policy must be Best Effort (not Reliable)

■ Jetson gets ~2.5 Hz pointcloud on home WiFi vs ~0.4 Hz on PortentaROS hotspot

12. Project Owner

Field	Details
Name	Ricardo Manriquez
Role	Senior Data Analyst — Peterson CAT, San Leandro CA
Education	UC Berkeley MAS-E — AI/ML, Robotics & Controls (GPA 3.9, Expected Dec 2026)
GitHub	github.com/RicardoMan
LinkedIn	linkedin.com/in/ricardomanriquez
Demo Video	https://youtu.be/PtZjn18mQqo
Location	Oakland, CA (originally from Chile)